

Reusable, cross-species-validated pipeline for tissue-specific AAV regulatory elements

One pipeline · any cell type · only the argument changes — a synthetic-biology “parts registry” for gene-therapy design

WHY IT MATTERS

- Tissue-restricted expression → lower off-target toxicity, dose & immune load
- Cross-species failure is the field’s costliest failure mode: elements tuned in mouse routinely fail in human/NHP
- We flag likely non-translating parts computationally, up front

REUSABILITY IS STRUCTURAL, NOT CLAIMED

```
$ python run.py "Pancreatic_Beta_Cell"
$ python run.py "Small_Intestinal_Enterocyte"
```

Code lines changed between the two runs:

0

Only the cell-type string differs. No per-tissue branching.

OPEN DATA

CATLAS scATAC (1.14M cCRE × 222 cell types) · JASPAR · GENCODE
hg38 · CELLxGENE Census · Ensembl orthologs | code: MIT

THE FUNNEL (same thresholds, both tissues)



~1.1M candidate elements → a ranked, translation-aware shortlist per tissue

POSITIVE CONTROL ✓

Top beta parts at INS (rank 1) + IAPP
Enterocyte: OLFM4 / APOA4 / PIGR / RBP2
Housekeeping (ACTB...) → $\tau \approx 0.02$, filtered
Strict mode: 25/25 beta parts gene-linked

MOTIF ENRICHMENT ✓

Beta shortlist 4.7× enriched
for PDX1/NKX6-1/PAX6/NEUROD1
 $p = 1.6 \times 10^{-13}$ (72% vs 12% PDX1)
Identity-TF PWMs from JASPAR, per tissue

CROSS-SPECIES ★ headline

Every part flagged for translatability
Enterocyte PHGR1 → no mouse ortholog:
human-specific; a mouse-only run misses it
MALRD1 → divergent mouse expression